Effect of Co/Ni codoped ZnO Nanoparticles on Structural, Optical and Magnetic Properties

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Abstract - We report systematic studies of the magnetic properties of Co and Ni co-doped ZnO nanoparticles prepared by co-precipitation method. Powder diffraction reveals that Co and Ni ions substituted into ZnO lattices without any secondary phase formation. Photoluminescence shows that the Co/Ni were doped into the ZnO lattice resulting slight shift in near-band-edge emission and the presence of donor defects and oxygen vacancies in the prepared samples observed in Raman spectra. Magnetic measurements of the nanoparticles exhibits ferromagnetic behavior at room-temperature.

Keywords: Nanoparticles, Powder diffraction, Photoluminescence.

I. INTRODUCTION

In recent years, large efforts have been devoted to the investigation of DMS due to their potential applications in spintronic devices for which the basic requirements are room temperature ferromagnetism (RTFM) and good optical transmittance. Zinc oxide (ZnO) is an attractive semiconductor for numerous applications and a building block material for many advanced device technologies because of its large band gap (3.35 eV) at room temperature. It has large excitonic binding energy (60 meV), optical transparency, chemical stability, hardness and piezoelectric properties [1]. For semiconductors, doping of metal elements is a powerful tool to tailor the electrical and optical properties for the fabrication of optoelectronic devices. More interestingly, the results from previous theoretical and experimental studies have shown that ZnO doped with appropriate transition metals were considered as diluted magnetic semiconductors, which attracted a great deal of interest due to their ferromagnetism at room temperature. They can change the microstructure and the optical properties of ZnO system. Many synthesis methods were employed for the preparation of TM doped ZnO nanocrystals. The co-precipitation is one of the most important methods to prepare the nanoparticles in large scale with low cost. Some reported works on TM co-doped ZnO system. Therefore, in the present investigation, $Zn_{0.80}Co_{0.10}Ni_{0.10}O$ nanoparticles are synthesized and the effect of Ni dopant on its structural, optical and magnetical properties were studied and discussed in detail.

II. SYNTHESIS

Synthesis of pure, Co doped ZnO, Ni doped ZnO and Ni co-doped Co/ZnO was carried out using analytical grade zinc acetate dihydrate [Zn $(CH_3COO)_2.2H_2O$], cobalt acetate dihydrate [Co $(CH_3COO)_2.2H_2O$], nickel acetate dihydrate [Ni $(CH_3COO)_2.2H_2O$] and sodium hydroxide (NaOH) in as-received condition. In the synthesis process, a required amount of zinc acetate was completely dissolved in deionized water and a required amount of aqueous NaOH solution was added drop wise to the mixture. Later, the solution was stirred at room temperature for 30 min, and then kept at 80°C for 5 hours. Thereafter, white precipitate was formed and it was washed several times with double distilled water and ethanol. Finally, the precipitate was dried at 120°C for one hour. The prepared samples were annealed at 400 °C in air for 2 hours. Thus, the nanopowder of ZnO NPs was obtained and used for further studies.

For the synthesis of $Zn_{0.80}Co_{0.20}O$ (Co/ZnO) nanoparticles, the calculated amount of cobalt acetate in water was mixed with an aqueous zinc acetate solution. The required amount of aqueous NaOH solution was added drop by drop to the above homogenous mixture to get a white precipitate with a pale pink color. The same procedure adopted for the preparation of Co doped ZnO was followed for $Zn_{0.80}Ni_{0.20}O$ (Ni/ZnO) and $Zn_{0.80}Co_{0.10}Ni_{0.10}O$ (Ni-Co/ZnO) NPs. Ni doped ZnO and Ni co-doped Co/ZnO NPs were obtained from a homogenous mixture to get a precipitate with a pale green color and black color, respectively.

III. Results and Discussion

Figure 1 shows XRD patterns for all the samples which are found to be in good agreement with the standard peak positions of pure ZnO (JCPDS card no. 36-1451); no secondary peaks from dopant ions or metal oxides such as CoO, Co_2O_3 or NiO was found. The synthesized sample shows the formation of nano crystalline powder and hexagonal wurtzite structure with space group P63mc. The replacement of Zn ions by Co and Ni ions decreases the intensity of XRD peaks, indicating that the degree of crystallinity of samples decreases, while the concentration of the defect increases.

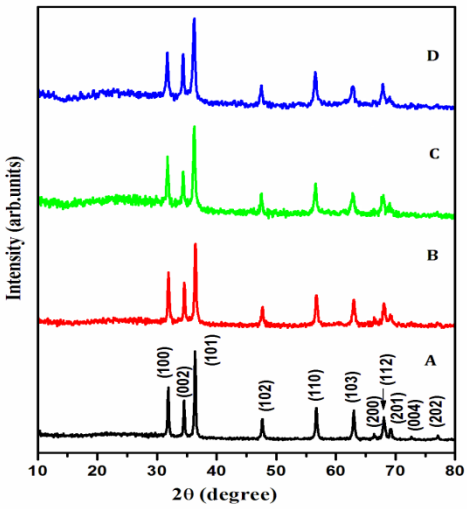

FIGURE 1. XRD patterns of (A) undoped ZnO, (B) $Zn_{0.80}Co_{0.20}O$, (C) $Zn_{0.80}Ni_{0.20}O$ and(D) $Zn_{0.80}Co_{0.10}Ni_{0.10}O$ NPs.

PL spectra of undoped and doped ZnO nanoparticles are shown in Figure 2 (a). The first deconvoluted peak around 391 nm corresponds to near band edge (NBE) emission which shifts into blue region with the doping of Ni and Co ions. In general, oxygen vacancy (V_O) and zinc interstitial (Z_{ni}) are donors, while zinc vacancy (V_{Zn}) and interstitial oxygen (O_i) are acceptors in ZnO [2]. Among these point defects, only Z_{ni} is a shallow donor and the corresponding defect level could be located slightly below the conduction band edge. In Pure ZnO, blue emission around 407, 437 and 490 nm can be attributed to the transition from extended Z_{ni} states, which are slightly below the simple Z_{ni} state to the valence band. The green emission exhibits at 520 nm due to the single ionized oxygen vacancy and this emission was originated from the recombination of a photo-generated hole with the single ionized charge state.

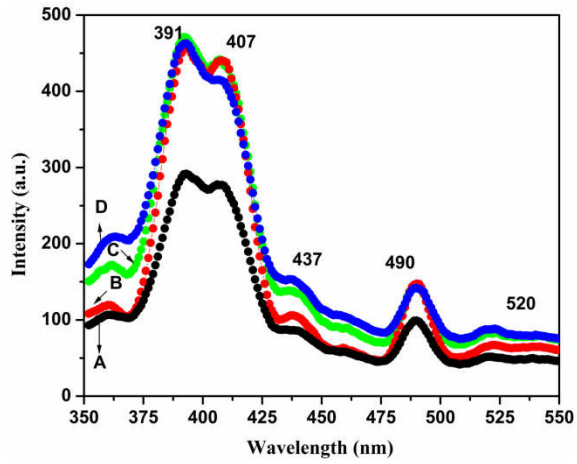

FIGURE 2. PL spectra of (A) undoped ZnO, (B) $Zn_{0.80}Co_{0.20}O$, (C) $Zn_{0.80}Ni_{0.20}O$ and(D) $Zn_{0.80}Co_{0.10}Ni_{0.10}O$ NPs.

The blue and green emission bands are originated from the intrinsic defects and donor-acceptor pair recombination involving Co and Ni impurity acceptors. Green emission bands at 520 and 532 nm arises from the oxygen vacancies in Zn-Co-Ni-O lattices due to the increase in the impurity levels in the samples. Hence, the co-doping tends to increase the distortion centers in the lattice of the samples. In addition, a slight shift of UV peaks towards the lower wavelength side is found with the substitution of Ni^{2+}, Co^{2+} ions. This can be attributed to the blue shift of energy gap with co-doping of Ni/Co ions. The occurrence of blue shift in the UV emission peak is due to the Burstein-Moss effect.

The room temperature Raman spectra were recorded for pure and doped samples with the frequency range from 100 to 800 cm^{-1} as shown in Figure 3. The mode observed at 429 cm^{-1} can be assigned to the high frequency branch of E_2H mode, which involves oxygen motion characteristic of wurtzite structure [3]. The presence of E_2H mode in all the four samples indicates the hexagonal wurtzite structures, which are consistence with the XRD studies.

In comparison to the Raman spectrum of undoped ZnO, the additional mode exhibits at 525 cm^{-1} for Co, Ni and Ni co-doped Co/ZnO samples can be assigned to the quasi-LO phonon mode due to the abundant shallow donor defects such as zinc interstitials Z_{ni} and/or oxygen vacancies V_O, bounded on the tetrahedral Co/Ni sites. The existence of the quasi-LO phonon mode manifests the incorporation of Ni/Co in the ZnO lattice. The shift of E_1 (LO) mode confirms the presence of defects (oxygen vacancies) in Co, Ni and Co-Ni co-doped ZnO crystalline samples.

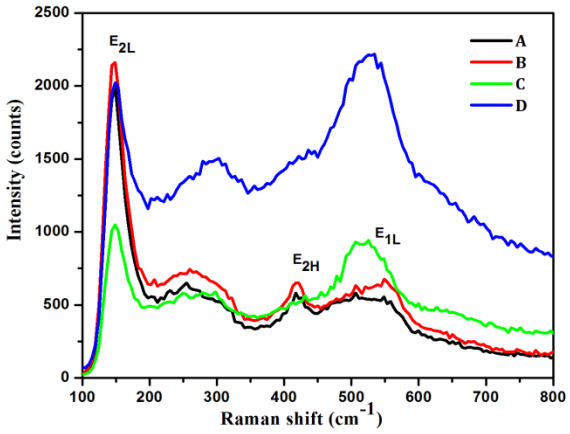

FIGURE 3. Micro-Raman spectra of (A) undoped ZnO (B) $Zn_{0.80}Co_{0.20}O$, (C)$Zn_{0.80}Ni_{0.20}O$ and (D) $Zn_{0.80}Co_{0.10}Ni_{0.10}O$ NPs.

The undoped ZnO sample exhibits diamagnetic behavior. The FM behavior can be attributed to the incorporation of Ni and Co ions in ZnO structure. The room temperature ferromagnetism (RTFM) arises as a result of exchange interaction between delocalized charge carriers initiating to form oxygen vacancies and zinc interstitial in the crystal structure [4]. Further co-doping of Co content with Ni/ZnO sample increases the oxygen vacancy concentration that eventually changes the magnetic ordering of the sample which shows increased magnetization than other samples. RTFM observed in TM-doped ZnO nanoparticles is originated from two factors; i) the number of defects and oxygen vacancies; ii) the exchange interactions between the TM ions and the O ion spin moment . The existence of ferromagnetism in $Zn_{0.80}Co_{0.20}O$ and $Zn_{0.80}Ni_{0.20}O$ may be due to the clustering of metallic Co/Ni and/or intrinsic ferromagnetism from the charge carriers. XRD results clearly indicate the absence of metallic Co and Ni phase in the sample and therefore the observed ferromagnetism at room temperature is an intrinsic property of Co and Ni doped ZnO. It is believed that the existence of ferromagnetism in Co:ZnO and Ni:ZnO could be due to the exchange coupling between localized 'd ' spins on the Co and Ni ions mediated by delocalized carriers [5]. When Ni was co-doped in $Zn_{0.90}Co_{0.10}O$, the magnetic behavior is found to change in the sample. The incorporation of the element Ni is conducive to the formation of FM order. The changes in the magnetic behavior can be attributed to the co-doped Ni which would create additional carriers. In addition, the defects present in the samples are also expected to enhance magnetic behavior in Co doped ZnO samples than in Ni doped ZnO. According to the RKKY theory, the ferromagnetic interactions in DMS materials are usually produced due to the exchange interaction between local spin-polarized electrons (such as the electrons of Co^{2+} and Ni^{2+} ions) and conductive electrons or free carriers.

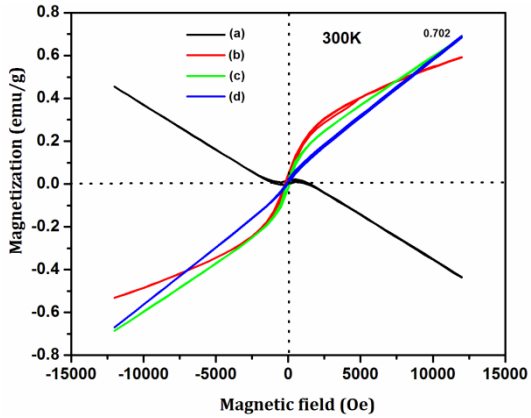

FIGURE 4. Magnetic behavior of (a) undoped ZnO, (b) $Zn_{0.80}Co_{0.20}O$, (c) $Zn_{0.80}Ni_{0.20}O$ and (d) $Zn_{0.80}Co_{0.10}Ni_{0.10}O$ NPs.

IV. Conclusion

Magnetic properties of undoped and doped ZnO nanoparticles synthesized by co-precipitation method are investigated. X-ray diffraction and Raman measurements indicate the wurtzite structure for undoped and doped ZnO with substitution of Co^{2+} and Ni^{2+} in host lattice sites. Co, Ni doped ZnO and Ni co-doped Co/ZnO nanoparticles have been successfully prepared by co-precipitation method. A dominant UV emission corresponding to the recombination of excited electrons from the valence band with the holes in the conduction band and defect related blue and green emissions are observed. Raman studies confirmed the presence of oxygen vacancies which are believed to promote the RTFM in the grown ZnO nanoparticles. RTFM was observed for the doped samples which may be due to the creation of free carriers and hence it is expected to be considered as a promising material for spintronic applications.

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